

HUSNA FATHIMA

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PROFESSIONAL SUMMARY

Computer Science undergraduate (VTU) specializing in IoT, Cybersecurity, and Blockchain, with a builder's portfolio spanning intrusion detection, cloud-based threat monitoring, smart contract security, and UAV cyber-resilience research. Combines hands-on security engineering with a research-driven mindset and proven leadership founding India's first student-powered DevRel community. Currently developing SCOUT-X, an onboard UAV threat-response framework, for IEEE publication. Looking to bring the same rigor and ownership to a Security/Software Engineering role.

EDUCATION

Bachelor of Engineering — Computer Science & Engineering — K.S. Institute of Technology, Bengaluru (VTU) *Sep 2023 – 2027*

Specialization: IoT, Cybersecurity, and Blockchain Technology

EXPERIENCE

Industry Connect Lead / Student Community Ambassador — Microsoft Student Community (AI NextGen) *Apr 2025 – Sep 2025*

- Co-founded and established India's first student-powered DevRel squad to empower aspiring developers through structured community programs.
- Built and maintained relationships with key community stakeholders, driving collaborative outreach events and partnerships.
- Engaged community members through targeted outreach initiatives, strengthening long-term participation and connection.

RESEARCH & PUBLICATIONS

SCOUT-X: Secure Command and Operations for UAV Telemetry — Exfiltration Defense *In Progress · Targeting IEEE Submission*

- Designing an onboard Cyber Incident Response Engine (CIRE) enabling autonomous UAV threat detection, classification, and tiered response independent of ground-control intervention.
- Built telemetry simulation and a security middleware pipeline with trust-score based anomaly validation, GPS-speed consistency checks, and an attack classification engine for spoofing, hijack, and desynchronization scenarios.

PROJECTS

HashGuard — File Integrity Monitoring & Security Analytics System | *Python, Flask, SQLite, SHA-256, ReportLab*

- Built a Flask-based system detecting unauthorized file modifications, deletions, and additions using SHA-256 baseline hashing, with severity-classified event logging in SQLite.
- Delivered a live analytics dashboard, JSON/PDF/CSV report generation, and email alerting for critical security events, alongside secure authentication with password hashing.

CloudShield — Cloud Log Monitoring & Threat Detection | *Python, Flask, AWS S3, Boto3*

- Built a Flask application that parses uploaded log files to detect brute-force attempts, port scanning, unknown-IP access, and request flooding.
- Designed a modular backend that securely archives logs to AWS S3 and renders results on a threat-analysis dashboard.

Network Attack Pattern Analyzer | *Python, Scapy, Matplotlib*

- Built a CLI tool that analyzes PCAP traffic using Scapy to detect port scanning, traffic flooding, and SYN flood attacks.
- Implemented packet feature extraction, timestamped logging, and attack visualization with Matplotlib.

AuditForge — Smart Contract Security Analysis Platform | *Solidity*

- Building a static analysis platform to identify vulnerabilities in Solidity smart contracts.

TradeArena — Real-Time Stock Market Simulator | *React, Recharts, Node.js, Express, MongoDB*

- Built a full-stack trading simulator with live price updates, portfolio tracking, PnL calculation, and a leaderboard ranking system.
- Deployed frontend on Vercel and backend on Render, with MongoDB-backed persistence for user portfolios.

Chimera — Open-Source Local AI Platform | *Python, Ollama, RAG, Vector DB*

- Architected an extensible local-LLM platform combining long-term memory, Retrieval-Augmented Generation, tool execution, and autonomous agents.
- Designed a plugin-based architecture extending the platform to cybersecurity, blockchain, and IoT automation tooling.

IoT Dashboard | *React.js*

- Built a React-based front-end dashboard for real-time IoT device monitoring and visualization.

Stock Simulator | *Python, MySQL Workbench*

- Built a mock stock-market simulator for practicing day and night trading strategies, backed by a MySQL database.

SKILLS

Programming: Python, Java, C, JavaScript, SQL, Solidity

Frameworks & Cloud: Flask, React.js, Node.js/Express, MongoDB, AWS S3, Git/GitHub, Kubernetes, Jenkins, Linux

Cybersecurity: OWASP, CTF methodology, network reconnaissance, Incident Response, Vulnerability Assessment, scripting & automation

Languages: English (Advanced), Hindi (Advanced), Urdu (Native), German (Limited)

CERTIFICATIONS

- AWS Practitioner — Cloud Institution
- Research Methodology — NPTEL, IIT Madras
- Front End Web Development — IBM SkillsBuild PBL Program (CSRBOX)
- Java Data Structures & Algorithms — Infosys Springboard
- Problem Solving & SQL (Basic) — HackerRank
- AI Agent Summit, Bengaluru — Azure Developer Community × Reskilll
- Microsoft AI Skills Fest — Guinness World Record: Most Users in an Online Multi-Level AI Lesson (24 hrs)